On Induction and Verifiability of Reichenbach

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Abstract

In this paper, I will clarify the features of induction and attempt to address the major issue of inductive inference: is it possible to close the gap of the premises and the conclusion? With the help of Reichenbach's approach, it is possible to resolve the problem. According to Reichenbach, a sentence is meaningful because it can be verified. But what is verifiability? I will further analyze this concept based on Reichenbach's theory of verifiability.

Keywords: Reichenbach, Induction, Posit, Scientific philosophy, Verifiability

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論萊興巴赫的歸納法與可檢證性

賴 文*

摘要

本文首先釐清歸納法的特性，並嘗試處理歸納推理中的重大問題：有可能關閉前提與結論間的鴻溝嗎？以萊興巴赫的方法做為輔助，這個問題有可能被解決。根據萊興巴赫的看法，一個句子之所以有意義，是因為它可以被檢證。但什麼是可檢證性？我根據萊興巴赫的可檢證性理論，進一步分析這個概念。

關鍵詞：萊興巴赫、歸納法、假設、科學哲學、可檢證性

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Introduction

I intend to deal with the problems rooted in induction and verifiability. Even though these problems have long tradition, Reichenbach makes them revive.

The first problem is that there is always a gap between the collected premises and the conclusion in induction. However, Reichenbach offered us a concept named ‘posit’, once we treat the conclusion through inductive inference as a posit, the problem of induction seems to disappear. Following this line, I will start off with clarifying the problem of induction itself, then investigate the concept of ‘posit’ more deeply: what is posit? Does it really resolve the problem?

Secondly, an inductive inference is never isolated, it is always ready for justification and correction by other inferences according to Reichenbach. On this view, what is the fundamental reason for us to employ other inferences to justify or correct an inductive inference?

The final point is that a sentence is meaningful if and only if it is verifiable. This claim is arguably the central spirit of Reichenbach’s scientific philosophy. Nevertheless, what is the requirement of verifiability? In other words, if a sentence is verifiable, what condition it should meet?

Posit

Deduction’s conclusion is a true statement if and only if all premises are true and the argument is valid. However, this kind of inference will never produce the conclusion which can be used for prediction. Since the conclusion is already logically entailed in the premises. In this sense, there is no gap between premises and conclusion in deductive inference. In contrast, induction can give us some information about the future because the conclusion from the induction is claiming something more than the premises do. From the premises to the generalized conclusion, it seems that we jump to the conclusion in inductive inference.

Why is it so hard to close the chasm? To be more precise, it seems that

we presuppose something in induction so as to this chasm can never be closed.

First, we presuppose that the conclusion from inductive inference is towards future. In other words, this conclusion is not only an assertion which says something about present, but can be applied to the future events. As to Reichenbach, "the inductive inference is the instrument of predictive knowledge"[1]. The conclusion from inductive inference seemingly contains the predictive power in order for us to predict the future and guide our action. For instance:

P1: The first crow I observe is black.
•
•
•
P500: The 500th crow I observe is black.

C: All crows are black.

In this conclusion, we presuppose that it gives us some information about the future events: if I encounter another crow, it will be black. Here the issue is that our expectation on the conclusion is too high. Since this conclusion with predictive power also claims that (we presuppose) at any point in the future, the crow we observe will be black. The chasm appears because we only have limited evidence, nevertheless, we use the evidence and wish to reach a generalized conclusion. The sky seems to be a little bit too high in comparison to our evidence at hand.

Second, when we make an inductive inference, we presuppose that the conclusion is a true statement. The central point here is that before we actually 'know' (through empirical evidence or theoretical spectulation) whether the conclusion is true or false, we implicitly presuppose that the conclusion from induction is true. This is the reason that we can never close the chasm between

[1] Reichenbach, Hans, *The Rise of Scientific Philosophy,* Berkeley: University of California Press, 1951, p.229.

the premises and the conclusion with this mindset. We can use the induction of the color of crows (see above) as an example: according to its conclusion, at any point when we find a crow, it must be black. Moreover, the conclusion also entails that any unobserved crow is black. Here is another kind of example of induction:

P1: John hates Mary as he admitted.
P2: At the crime scene, we find the blood stain of John.
P3: The knife which causes the death of Mary has John's fingerprints.
...

C: John is the perpetrator of this crime.

I use this example in order to illustrate that when we presuppose that this conclusion is a true statement, the chasm appears. There is always the possibility that John is not the perpetrator of this crime. For instance, the perpetrator of this crime is John's twin-brother Peter, or an alien sends a cloning-John to kill Mary (even though it is ridiculous, it is still possible). The point here is that we always have limited evidence and presuppose that the conclusion is a true statement, however, there is always the possibility that we can find empirical evidence to generate a different or contradictory conclusion. Therefore, the chasm persists because we presuppose that the conclusion is a true statement.

At this point, we need to know that a valid induction can have a false conclusion. Yet, the essential point here is that the chasm in induction can always have its place because before we know the truth-value of a conclusion from induction, we presuppose that this conclusion is a true statement.

In the light of Reichenbach, if we treat the conclusion of induction from a different perspective, then we can close this chasm. If we treat the conclusion as a posit, which means that based on the evidence (premises) we have, then we can have this conclusion which makes no contradiction to the premises and we could predict some future events from this inductive inference. This

conclusion might be proved to be false at a later time, but with this conclusion at present from induction, we can predict some phenomena in the real world. Reichenbach removes our presupposition that a conclusion from induction is a true statement, and therefore set out a relatively low and reasonable requirement for the conclusion from induction. In my interpretation, this shift is a successful and reasonable one because in fact, what we can have from these premises is just a posit, not a true statement which will be true at any point of time and will be held forever.

One of the solutions to close the chasm of induction is introduced by David Hume. He introduced Uniformity of Nature (UN) in order to justify the conclusion of induction. To wit, adding an extra premise as another pillar to reach the 'sky' (conclusion in induction treated as true statement). Nevertheless, this strategy will fail for three reasons according to my interpretation. The point is: it is likely that Hume himself knows this strategy does not work, his main purpose seems to show that when we make an induction, we already presuppose UN as a hidden premise.

The UN strategy can be illustrated as below:
P1: The first crow I observe is black.
•
•
•
P500: The 500th crow I observe is black.
P501: The unobserved will be similar to the observed.

C: All crows are black.

By introducing another premise, apparently, the gap between the premises and the conclusion is closed. Nonetheless, if we dig this concept more deeply, we will find that this strategy does not resolve the problem.

In the first place, when Hume introduces this extra premise, he cannot erase the possibility that we might find a crow which is not black. We can still

say that there is a possibility that in the future we will find a crow which is not black, to be more extreme, the scientist can create a not-black-crow in order to falsify the conclusion if the technology is mature enough. In short, UN cannot erase the possibility of new discovery so as to close the gap.

Another issue is that UN is a product of an induction. In other words, UN is a conclusion of another induction in order to become a premise of black-crow induction. It means that the procedure of producing UN is dubious since this procedure has a gap. In this sense, we cannot guarantee that UN is true. Moreover, how can we resolve the problem of a certain kind of argument by employing another conclusion produced by an argument of the same kind?

Finally, UN is not always true in nature. Intuitively and factually, we can find new cases which are not similar to the observed. The strategy of UN will certainly fail based on these reasons.

What is the main purpose of David Hume? In my view, he intends to show that when we make an induction which produces the generalized conclusion, we already add UN as a hidden premise. Put it in another way, human beings have the mental disposition of conceiving of nature in this way. When we make a conclusion from inductive inference towards the nature, we have the disposition to think that the unobserved will be similar to the observed. Second, in order to justify a conclusion, it is unreasonable to justify it by introducing another conclusion produced by the same kind of argument. In my opinion, Hume intends to show that we should not fall into this trap.

What is posit of induction in the sense of Reichenbach? "A posit is a statement which we treat as true although we do not know whether it is so."[2] Even if he did not give a precise definition of this term, we can treat it as a conclusion which is probably true, and therefore we can make prediction.

The concept of posit is the key to the understanding of predictive knowledge. A statement about the future cannot be uttered with the claim that

[2] Reichenbach, Hans, *The Rise of Scientific Philosophy,* Berkeley: University of California Press, 1951, p.240.

it is true; we can always imagine that the contrary will happen, and we have no guarantee that future experience will not present to us as real what is imagination today.[3]

If we find the new evidence that can falsify the posit, then we can modify it to be another conclusion which can involve the counter-example. For example:

P1: I throw this salt cube in the water, it dissolves.
P2: Peter throws a salt cube in the water, it dissolves.
...

C: Salt has the disposition of dissolving in the water.

We can predict that if a salt cube is thrown in the water, it will dissolve. However, if a salt cube does not dissolve in the water, how can we find the cause and modify our conclusion? For instance, we find that when a salt cube is thrown in the saturated water, it will not dissolve. We can then form the conclusion (posit) that "Salt has the disposition of dissolving in the water if and only if the water is not saturated". Here is another problem: if a posit contains ad hoc, when or to what extent should we abandon our original conclusion? If a posit has a bunch of ad hocs, it seems that this posit has some intrinsic problems. On this issue, it seems that we have no definitive standard to decide at which point we should abandon the posit. At best, we can only say that if a new theory has been proposed which faces less counter-examples and has same quantity of evidence comparing to the original one, at this point, we should accept the new theory.

However, a posit from certain inductions cannot play a role of explanation since this kind of explanation is not causal explanation. If Mary asks why Salt has this kind of disposition? We cannot answer her that because

[3] Reichenbach, Hans, *The Rise of Scientific Philosophy,* Berkeley: University of California Press, 1951, p.241.

every time we throw the salt cube in the water, it will dissolve. This repetition of a phenomenon cannot explain its cause. The central point is that repeatedly occurrences can justify a posit, but they cannot explain the mechanism of this posit being true.

The reliability (probability of being true) of a posit depends on the quantity of true premises in an induction. Still, in some cases, the increase of the quantity of premises do not increase the reliability of an conclusion. Consider the argument in religion:

P1: God creates the world.
P1: God lets the sky rain.
...

C: God exists.

In this case, we can find that no matter how much evidence we add into the space of premise, the probability of the conclusion being true will not go up. What is the problem here? In my interpretation, because we allow the indirect evidence to support the conclusion. Of greater importance, this kind of attribution (it rains because of God) is arbitrary, we just believe that the different phenomena are caused by the same existence without any justification. However, this kind of attribution and allowance for indirect evidence will cause some problems also in science, I will circle back to this topic in the third section.

Network

"In fact, practically all inductive inferences are made, not in isolation, but within a network of many inductions."[4] When we make an inductive inference, we produce the conclusions which can be used for prediction. Yet

[4] Reichenbach, Hans, *The Rise of Scientific Philosophy,* Berkeley: University of California Press, 1951, p.244.

this conclusion can and should be corrected by other conclusions in the same network. For specific conclusion (for example, the color of crows), it is hard to set the boundary to claim that which and how much information is contained in this network. In other words, we can never specify all information contained in a network, which is used to correct the specific conclusion. The boundary of a certain network is moveable, because of the development and discovery of science. To this extent, it is unlikely for philosopher to be concerned with the amount of information; what should be at issue is the question of justification.

Consider the following cases:

Argument A	Argument B
P1: The first crow I observe is black.	P1: Species x varies in color.
•	P2: Species y vaires in color.
•	
•	
P500: The 500th crow I observe is black.	
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C: All crows are black.	C: All species varies in color.

In this context, the conclusion from Argument A is subjected to the conclusion from Argument B. Hence, the first conclusion can be modified in this way: Most crows are black but some are not. This is an illustration of correction within a network. In this sense, we can claim that an inductive conclusion which is corrected by more conclusions can have higher probability of being true.

Here raises a question: is this correction a logical fallacy because we use a conclusion of induction to modify another conclusion of induction? I will give the reason to show that why this example will not fall into the trap of logical fallacy and why Hume's UN strategy does. Consider Hume's UN strategy:

P1: The first crow I observe is black.
•
•
•
P500: The 500th crow I observe is black
P501: The unobserved will be similar to the observed.

C: All crows are black.

Hume intends to close the chasm between the premises and the conclusion by adding an extra 'true statement', nevertheless, this premise (UN) is from an induction which has the same chasm and thus this premise is not a true statement. Additionally, in fact and by intuition, we can know that UN is not true without using induction. Still, in the example of crows, our main purpose is not to settle the problem of induction. Instead, we just introduce extra information to increase the credibility of original conclusion. In my interpretation, the essential difference is that in Hume's case, he tries to use inductive conclusion in order to resolve the problem of induction; namely, he uses a conclusion from an argument with flaw to prove that this kind of argument is flawless. Nevertheless, in Argument A, we just employ other information to correct our conclusion. In short, our intention is not to resolve the problem of induction per se but trying to increase the probability of being true in this conclusion.

Consider the following case:
P1: Angels have no extension.
P2: An angel can stand on a needle.
•
•
•

C: A needle can be stood by infinite numbers of angels.

The metaphysical theory's credibility cannot be increased or decreased by other conclusions in the same network. In other words, when one finds an inductive conclusion in order to correct the conclusion of angel's argument; the person who made the angel's argument can always claim that other people do not fully understand the characteristics of angel and so on in order to reject the correction. We can view angel's argument from a different perspective: no matter what evidence you find to correct the conclusion, this conclusion will not have higher or lower probability of being true.

Verifiability

In the light of Reichenbach, a sentence is meaningful because it can be verified. Reichenbach's main purpose was presumably intending to remove the metaphysical and religious theories from his scientific philosophy, to be more precise, from his network. If a sentence is meaningful, it can be used to refer to something in the world. The theory construed by this kind of sentences can play an essential role in causal explanation and prediction. However, the metaphysical and religious theories can also explain a phenomenon and make the predication, what is the fundamental difference between theories of metaphysics or religion and those of composing of meaningful sentences?

In my interpretation, their inner structures are very different. The metaphysical and religious theories' atomic sentences cannot be verified or falsify by the entity or state of affair in this world. Nevertheless, the theory composed of meaningful sentence in the sense of Reichenbach, the atomic sentences within this kind of theory can be verified or falsify by the entity of state of affair in our world. Even though the metaphysical and religious theories can always explain a phenomenon very well and always protect its wrong prediction (for example, a prophet says that tomorrow is the end of the world, when it does not happen, the prophet can still claim that because God has saved us and so forth), they do not count as causal explanation and prediction which is helpful for us.

My reason is that these explanations and predictions come from the unverifiable and unfalsifiable sentences, which means that they say nothing. A theory with verifiable sentences can offer the causal explanation and the prediction which are helpful for us. For example, why this toy is sinking into the water? One can say that because the density of this toy is greater than that of water. In this case, we can see that every term is verifiable, so that this theory offers us the physical mechanism of a phenomenon. With this explanation, we can use this theory to make the prediction. If a theory makes the right prediction, we can maintain the theory; if it fails to make the right prediction towards an event, at least we can trace back to the verifiable term and then try to fix the theory.

The next question is that what is verifiability? In my interpretation, direct verification must meet this prerequisite:

The evidence which can verify the sentence should be 'potentially' accessible to every human being who have normal cognitive faculty. It means that the evidence either can be perceived directly or can be observed with the appropriate instruments. In the case of "it snowed on Manhattan island on November 28, A.D. 4", it is verifiable because with the appropriate instruments (time machine), we can verify it is true or not. If the evidence is only available to certain kind of people with special ability or cognitive faculty, this private experience is only for these people. This private experience cannot share with others, hence, it will be unreasonable to say that this private experience stemmed from perceiving the evidence can be used to verify the sentence.

Conclusion

By introducing the concept of posit, the gap in induction seems to be closed. This is a reasonable solution because originally, we presuppose the conclusion in induction is a true statement. This presupposition is

inappropriate because our premises do not give us sufficient evidence to believe that the conclusion is a true statement. A posit, instead, lower our expectation. Thus, the problem in induction can be resolved.

In a network, correcting a conclusion of induction through another inductive inference can increase the possibility of being true for the original conclusion. The point is that an induction is never isolated, it is always ready for correction.

Finally in my view, direct verification requires this prerequisite: accessibility.

References

Reichenbach, Hans, *The Rise of Scientific Philosophy,* Berkeley: University of California